

Factoring Cubes and Higher-Degree Expressions

Sum and difference of cubes, factoring by grouping, quartics in quadratic form, and reading zeros off the factors

How to work:

- **Always pull the GCF first.** What is left is usually a pattern you know.
- **Cubes:** $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ and $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. The binomial keeps the sign of the original; the trinomial's middle term takes the opposite sign, and its last term is never negative.
- **Grouping:** four terms, two at a time — pull a GCF out of each pair and hope the leftover binomials match.
- **Quadratic form:** if the exponents are $2n$ and n , treat x^n as a single letter, factor, then substitute back and keep factoring.
- Factor completely over the integers, and box your final factorisation.

n	1	2	3	4	5	6
n^3	1	8	27	64	125	216

Perfect cubes worth recognising on sight.

1. Difference of cubes. Factor completely: $x^3 - 8$.

Name a and b first, then write the two factors.

Answer: _____

2. Sum of cubes with coefficients. Factor completely: $27x^3 + 64$.

Both terms are perfect cubes even though neither is a bare power of x .

Answer: _____

3. Common factor first. Factor completely: $2x^3 - 54$.

Neither term is a perfect cube as written. Pull the greatest common factor out before you look for the pattern.

Answer: _____

4. Factor by grouping. Factor completely: $x^3 + 3x^2 - 4x - 12$.

Group the first two terms and the last two, pull a common factor out of each pair, then factor what remains — the leftover quadratic still factors.

Answer: _____

5. Quadratic in disguise. Factor completely: $x^4 - 13x^2 + 36$.

Let $u = x^2$. Factor the quadratic in u , substitute x^2 back, and finish factoring each piece.

Answer: _____

6. From factors to zeros. The polynomial $P(x) = x^3 + 2x^2 - 9x - 18$ factors by grouping.

- Factor $P(x)$ completely.
- List all real zeros of P in increasing order.
- Complete the sign table below by writing $+$ or $-$ for $P(x)$ on each interval.

$x < -3$	$-3 < x < -2$	$-2 < x < 3$	$x > 3$

Answer: _____

7. Solve a quartic. Solve for all real x : $x^4 - 5x^2 + 4 = 0$.

Factor into four linear factors, then apply the zero-product property. Write the solutions in increasing order.

Answer: _____

8. Find the mistake. Sam factors $8x^6 - 27$ and writes

$$8x^6 - 27 = (2x^2 - 3)(4x^4 - 6x^2 + 9)$$

- Explain which part of the difference-of-cubes pattern Sam broke.
- Write the correct complete factorisation.

Answer: _____

9. Zeros and shape. Let $Q(x) = x^3 - 4x^2 - 4x + 16$.

- Factor $Q(x)$ by grouping and list its real zeros in increasing order.
- Sketch the graph of Q on the axes below: mark the zeros, get the end behaviour right, and make the curve sit on the correct side of the x -axis between them. An accurate shape is enough — exact turning points are not needed.



Answer: _____

10. Everything at once. Let $R(x) = 3x^5 - 24x^2$.

- Factor $R(x)$ completely over the integers. (GCF first, then a cube pattern.)
- List the real zeros of R , and say which factor produces each one.
- One quadratic factor contributes no real zero. Which one, and how can you tell without solving it?

Answer: _____